
Real-World PBTs as Lean Challenges

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Abstract

We present a benchmark for evaluating AI models and agents on real-world formal software verification tasks. We first scrape $\approx 20k$ property-based tests (PBTs) from permissively licensed real-world Python repositories, then automatically translate them into Lean 4 specifications with sorry placeholders. Translating PBTs into Lean specifications is challenging: it requires modeling Python semantics in Lean, inferring the logical property encoded in an imperative PBT, and handling the inherent difficulties of dependently-typed programming in a seldom-used language. We describe a three-agent LLM pipeline for transpiling PBTs into Lean specifications, evaluate coverage and quality metrics, and provide baselines for proof generation using several automated and model-based approaches. Our benchmark aims to drive progress on the underexplored problem of AI-assisted formal verification of real-world software, which we claim is of increasing importance as AI produces more and more of the world’s code.

1 Introduction

Several AI safety proposals are premised on implementing safeguards by way of *formal verification* (FV)¹ – that is, software systems that can produce and unequivocally certify mathematical proofs. These systems can be split into *interactive theorem provers* (ITPs) which accept and validate user-supplied proofs, and *automated theorem provers* (ATPs) which attempt (but often fail) to prove theorems fully autonomously. ARIA Safeguarded AI [8], Guaranteed Safe AI [7], and Scalable Formal Oversight [40, 29], among others, all suggest that in the future, all AI-generated software will be formally verified by autonomous FV agents, who will steer ITPs in much the same way that coding agents currently steer software production.

The only way FV can play this role is if agents can conduct verification work at least as well and as quickly as they can produce code. Otherwise, FV becomes a bottleneck and will naturally wither under the capital pressure to iterate². There is currently little evidence that such capabilities are possible. To achieve them, we would first need high-quality benchmarks to train on. Existing benchmarks (see Table 1) are too small [25], focused on advanced mathematics rather than engineering [14], or derived from expert-driven verification projects [6] which are quite different from what you would get were you to verify the average software project. To clarify this last point: historically, FV has been bottlenecked by the number of human expert practitioners able to write proofs in an ITP³, and therefore, FV was only ever applied to the most mathematically complex and safety-critical

¹FV forms a subset of *formal methods*, which also include PBT, model checking, and abstract interpretation.

²The reader is referred to [33] for a detailed discussion of how socioeconomic pressures degrade safety cultures and cause predictable accidents.

³We were unable to find any formal estimate of the number of human formal methods practitioners today. However, as a very rough proxy, de Almeida Borges et al. [9] conducted the largest ever survey of users of any particular ITP – in this case Coq, which is now called Rocq – at 466 participants. As a point of contrast, that year’s survey by the Rust foundation of Rust software engineers had 7,156 participants [36]. Anecdotally, most ITP work today is in Rocq, Lean 4, or Isabelle.

applications, such as compilers [22] or an OS kernel [20]. These projects are generally monolithic, closed systems, in contrast to the average software project which may have a frontend and backend, a database, reliance on several external black-box APIs such as an inference provider or a logging service, and an unnecessarily large dependency tree. The latter category of (normal) software is actually in many ways much more challenging to verify because it necessitates reasoning about black-box external systems and the interactions of several different components, potentially written in different languages and operating on different chips [4, 31].

MvH: A reviewer is going to ask us how we handle all this complexity. Do we have theorems that relate to external APIs? In what ways do we REALLY differ from other benchmarks? We need to substantiate this.

Benchmark	Lean	Isabelle	Coq/Rocq	HOL Light	Metamath	Dafny	Verus	Theorems	Focus
Minif2f [44]	✓	✓		✓	✓			488	math Olympiad
PutnamBench [39]	✓	✓	✓					640	math Olympiad
Fimo [24]	✓							149	math Olympiad
DafnyBench [25]						✓		782	scraped Dafny proofs
ProofNet [1]	✓							371	U math
RLM25 [32]	✓							619	G math
Lean Workbook [42]	✓							57,000	HS, U, G math
FormalMATH [43]	✓							5,560	math Olympiad
LEAN-GitHub [41]	✓							28,597	scraped Lean proofs
ProverBench [34]	✓							325	math Olympiad
MiniCTX [17]	✓							762	six Lean projects
LeanGeo [35]	✓							122	HS, U, G math
IneqMath [27]	✓							1,522	math Olympiad
FATE- $\{M,H,X\}$ [19]	✓							350	U, G math
VeriBench [3]	✓							857	software verification
CLEVER [37]	✓							161	software synthesis & verification
Vericoding [5]	✓					✓	✓	12,504	software synthesis & verification
MUSTARD [18]	✓							5,866	HS, U math
FormL4 [26]	✓							14,510	U math
CoqStoq (Rango) [38]			✓					10,396	software verification
MSC-180 [23]	✓							180	U, G math
HOList [2]				✓				29,462	U, G math

Table 1: **Prior ITP benchmarks.** HS = high-school, U = undergraduate, G = graduate. If one benchmark improves upon another, we exclude the lesser. 3 indicates the proof assistants each benchmark targets.

In light of the limitations of prior works, we generate a new benchmark, FVSpec, from public uses of property-based testing (PBT). While FV itself is rarely used in software development, PBT is a “close sibling” technology which has seen significant adoption. In both FV and PBT, engineers write *specifications*: logical properties that the software must obey. For example, we might require that a Python function `isort(lst)` always returns a sorted permutation of its input `lst`. The difference lies in how this property is checked. In FV, the property is proved using mathematical techniques—this results in almost-perfect confidence, but requires proof engineering by highly expert teams [12]. In PBT, the code is randomly tested—e.g. for random values of `x`. This is a push-button process with no proof engineering, and provides only statistical assurance.

PBT is typically much cheaper than FV, and as a result has seen wider adoption. In fact, PBTs are typically seen as the first step toward FV. For example, in the interactive theorem prover ACL2s [11], any unproven theorem statement is automatically treated as a PBT until the author supplies a proof. Thus the idea of translating PBTs to theorems for FV is natural and, to an extent, industry-standard. We build on this. First, we scrape $\approx 20k$ permissively-licensed PBTs written in Python’s Hypothesis framework [28]. Each PBT in our dataset can be seen as a formal conjecture about a piece of Python

code, which the author would like to either substantiate (prove) or disprove. To make it possible for an AI to resolve these conjectures, we translate both code and PBTs to the theorem prover Lean, which is becoming the de-facto standard in AI formal verification. Once the Lean version of the conjecture is (eventually) proven, it can be called a *theorem*. Until then, it has a `sorry` placeholder where the proof belongs. This approach was pioneered by [13] on the FVAPPS verification benchmark. In FVAPPS, source programs were taken from the APPS benchmark by [16], translated to PBTs, and then lifted to theorems. We apply the same process at scale to PBTs found in the wild.

We use AI agents to translate from Python to Lean, in the style of FVAPPS. This process is necessarily lossy, that is to say, it does not guarantee strict equivalence. However, strict equivalence is only needed if the goal is to aid the authors of the original software by resolving their open conjectures. Although when applicable we consider this goal a positive secondary contribution of our work, our actual motivation is to build a large benchmark of realistic and high-quality software verification exercises, representative of the diversity of code found in “normal” codebases. Thus, we don’t worry whether or not our translations are perfectly equivalent. The result is tens of thousands of verification challenges. Each consists of four artifacts (Figure 1): (1) the original Python implementation, (2) the original Python PBT exercising it, (3) a complete Lean implementation of the same function (no `sorry`), and (4) a Lean theorem statement of the property with a `sorry` for the proof. Unlike existing FV benchmarks, each problem corresponds to properties of real-world software written by engineers with no guaranteed formal verification experience.

MvH: TODO: Summary of results.

The rest of the paper is organized as follows. We formalize the task in Section 2. We characterize our input corpus, detailing how we scraped and deduplicated $\approx 20k$ permissively-licensed Hypothesis tests from open-source repositories, in Section ??, and discuss how that corpus compares to a similar one released by the Hypothesis team [10]. Then, we describe our generation pipeline—scraping, dependency extraction, LLM-based transpilation with iterative Lean LSP repair, and post-production cleaning—in Section ?. We present baseline results in Section ?, discuss threats to validity in Section ?, survey related work in Section ?, and conclude in Section ?.

2 Task Definition

Each problem in the FVSpec benchmark consists of four components: (1) the original Python implementation of the function under test, (2) the original Python PBT exercising that implementation, (3) a Lean implementation of the same function—a fully defined, computable definition, no `sorry`—and (4) a Lean theorem corresponding to the PBT, with a `sorry` placeholder where the proof should go. The AI’s task is to fill in every theorem `sorry` with a Lean proof term that compiles cleanly under `lake build`; submissions are then scored by whether the proof retains any `sorry`, `admit`, or axioms beyond Mathlib’s defaults. The Lean kernel is the sole arbiter—we perform no separate semantic comparison against the original Python. Each problem also includes a URL pointing to the original source code and license information.

As a running example, consider `isort(lst)`. Figure 1 shows all four artifacts of a single FVSpec problem. The left column holds the original Python implementation (top) and the PBT exercising it (bottom); the right column holds the corresponding `Impl.lean` (top), a complete Lean port of the function, and `Spec.lean` (bottom), containing two theorems—one for sortedness, one for permutation—each with their own `sorry`. The AI’s task is to discharge both Lean `sorry` placeholders.

We encounter two practical issues when translating Python PBTs into Lean implementations and (unproven) specifications.

Nonsensical theorems. Our pipeline can emit well-formed but useless statements in three ways.

(i) *Trivial Unit-typed conjectures.* When the formalization agent cannot recover a function’s true return type, it falls back on `Unit` as a placeholder. The resulting theorem is a tautology—e.g. for our running example, `theorem isort_check : isort lst = () := by plausible`, which holds for any total function into `Unit` and so carries no verification value.

(ii) *Ill-typed or unresolved-symbol theorems.* The formalization agent works from the Python source and writes Lean by analogy. When it invents a Lean name that does not exist—for in-

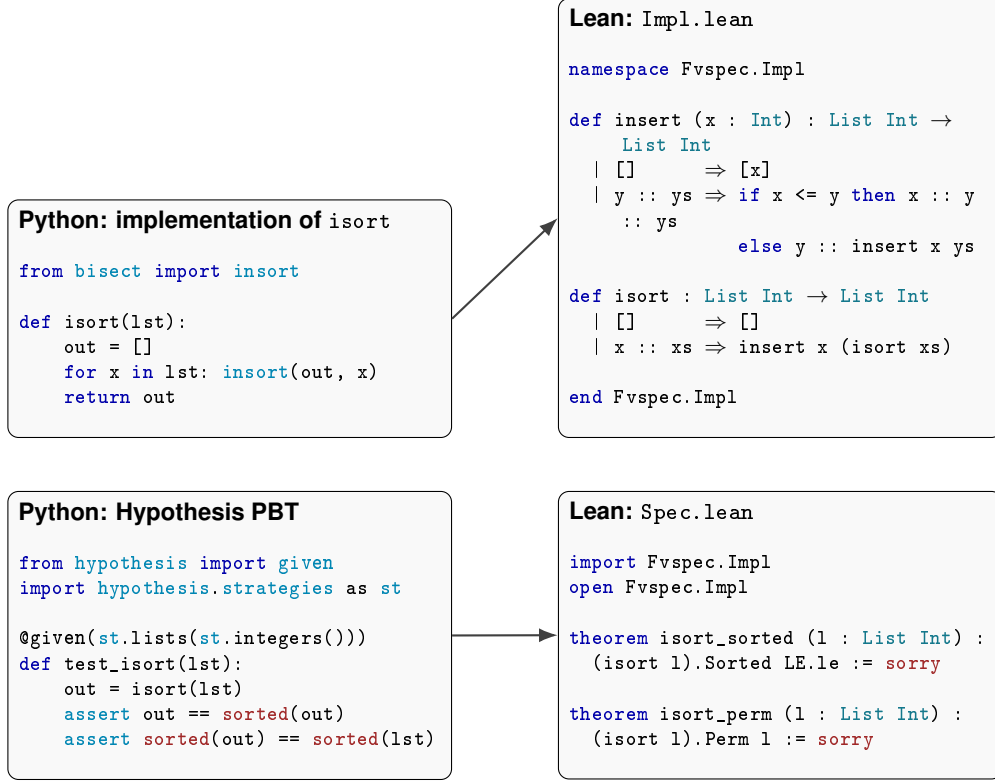


Figure 1: The four artifacts of a single FVSpec problem. Top: the Python implementation of `isort` and its complete Lean port. Bottom: the Hypothesis PBT and the corresponding Lean theorems. The AI must replace both Lean `sorry` placeholders—one per theorem.

stance, writing theorem `isort_perm (l : List Int) : (isort l).permutationOf l := sorry`, where `permutationOf` is not a member of `List` (the correct name is `List.Perm`)—or attaches a type signature inconsistent with the surrounding code, the Lean elaborator rejects the file with errors like *unknown identifier*, *type mismatch*, *function expected*, or *failed to synthesize*.

(iii) *Spec theorems leaking into Impl.lean*. Our pipeline runs two separate agents: one writes `Impl.lean` (definitions only), the other writes `Spec.lean` (theorems only). The implementation agent is explicitly instructed to emit no theorems, but occasionally—when shown the `isort` source—it pre-empts the spec agent by hallucinating a namespace `Fvspec.Spec ... theorem isort_sorted ... end Fvspec.Spec` block inside `Impl.lean`, which violates our file split and would silently expand the proof obligation if left in.

We catch (i) by regex (`detect_trivial_unit_theorems` in `lean_parsing.py`), drop (ii) at compile time (`validate/compiler.py`), and strip (iii) with a namespace-aware filter (`impl/lean_merger.py`). A theorem that survives all three filters is a useful challenge problem even if it has drifted from the original PBT—because Lean checks proofs against *the theorem we generated*, not the informal property.

Lean as a (nearly) perfect oracle. The Lean kernel arbitrates correctness, which is qualitatively much stronger than statistical sampling. It is not, however, adversarially perfect: an AI can satisfy the kernel by *changing the question*—adding the Axiom of Choice, weakening hypotheses, or exploiting a security vulnerability in Lean itself [15]—and the proof will still type-check [30]. To make gaming visible, every submission is scored on two axes: a binary *proved* flag (the file compiles under `lake build` and every original `sorry` has been discharged) and partial credit (fraction of original `sorry`s discharged). We additionally record any axiom dependencies the proof introduces, since an AI can trivialize a goal by quietly importing strong axioms; we do not currently auto-fail on extra axioms, but report them so reviewers can flag suspicious solutions. For higher-stakes evaluations, submissions

can additionally be re-checked through `leanprover/comparator` [21], which sandboxes the build and cross-validates the proof term against an independently-implemented Rust kernel (`nanoda`)—we do not require this today, but it is available if a future use case demands a stronger trust story. A sufficiently motivated AI can still smuggle a vacuous proof past us when the original theorem statement is itself too weak—a limit shared by every benchmark in this space.

Benchmarking FV is unusually hard: public verification artifacts are dominated by pedagogical exercises whose proofs already exist in many forms in pretraining corpora, the substantive industrial work (Intel, Synopsys, AWS, etc.) is closely held, and what remains is fragmented across tools (SAW/Cryptol, ACL2, Coq, Lean, TLA+) and small. We sidestep this by translating from PBT, a “close sibling” domain with abundant non-academic source material that was never written for a theorem prover. As a result, most theorems in FVSpec have likely never been verified by anyone, and—as Table 1 shows—FVSpec is by a wide margin the largest software-verification benchmark in this space and the only one whose problems originate from real-world, non-academic code authored outside of any theorem-proving context.

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